3.2.3 C: Matrices

	Content
C1	Add, subtract and multiply conformable matrices; multiply a matrix by a scalar.

	Content
C2	Understand and use zero and identity matrices.

C3	Use matrices to represent linear transformations in 2D; successive transformations; single transformations in 3D (3D transformations confined to reflection in one of $x = 0$, $y = 0$, $z = 0$ or rotation about one of the coordinate axes) (knowledge of 3D vectors is assumed).

	Content
C4	Find invariant points and lines for a linear transformation.

	Content
C5	Calculate determinants of 2 × 2 and 3 × 3 matrices and interpret as scale factors, including the effect on orientation.

	Content
C6	Understand and use singular and non-singular matrices; properties of inverse matrices. Calculate and use the inverse of non-singular 2 × 2 matrices and 3 × 3 matrices.

	Content
C7	Solve three linear simultaneous equations in three variables by use of the inverse matrix.

	Content
C8	Interpret geometrically the solution and failure of solution of three simultaneous linear equations.

	Content
C9	Factorisation of determinants using row and column operations.

	Content
C10	Find eigenvalues and eigenvectors of 2 × 2 and 3 × 3 matrices. Find and use the characteristic equation. Understand the geometrical significance of eigenvalues and eigenvectors.

	Content
C11	Diagonalisation of matrices; $\mathbf{M} = \mathbf{U}\mathbf{D}\mathbf{U}^{-1}$; $\mathbf{M}^{n} = \mathbf{U}\mathbf{D}^{n}\mathbf{U}^{-1}$; when eigenvalues are real.